INTRODUCTORY BUSINESS STATISTICS

Chapter 7 THE CENTRAL LIMIT THEOREM

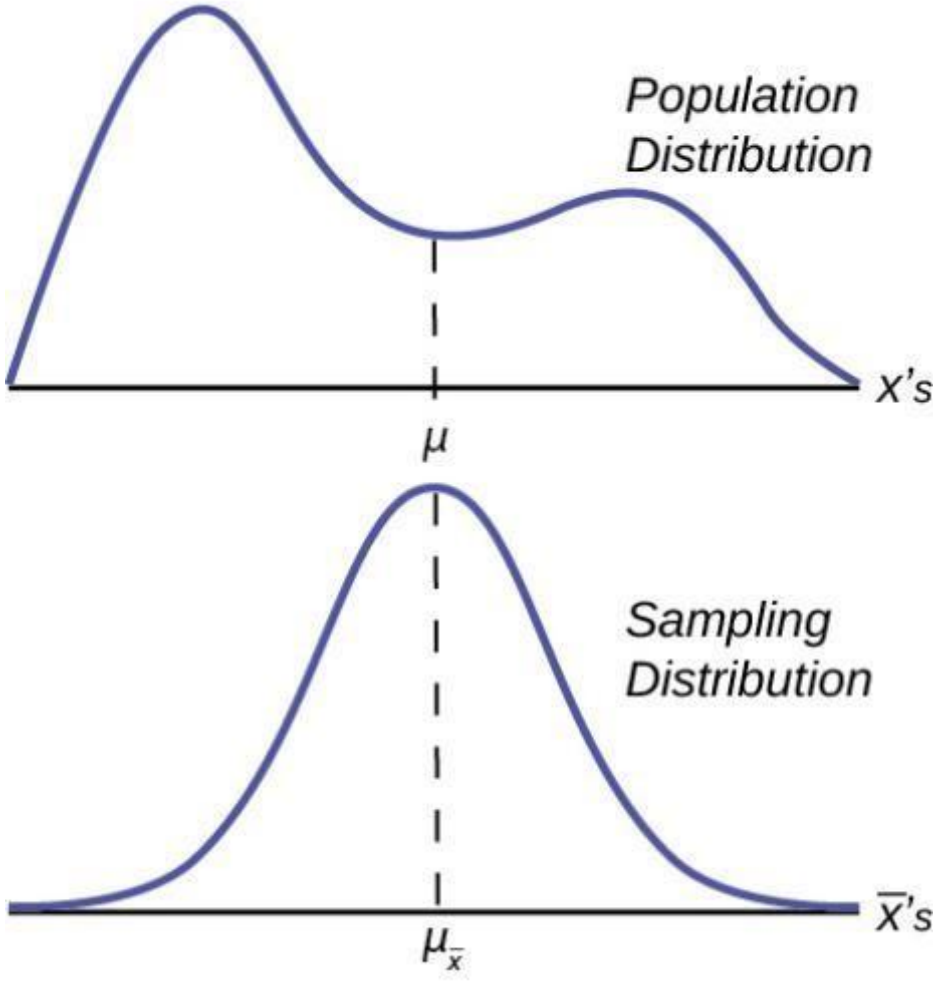


FIGURE 7.1

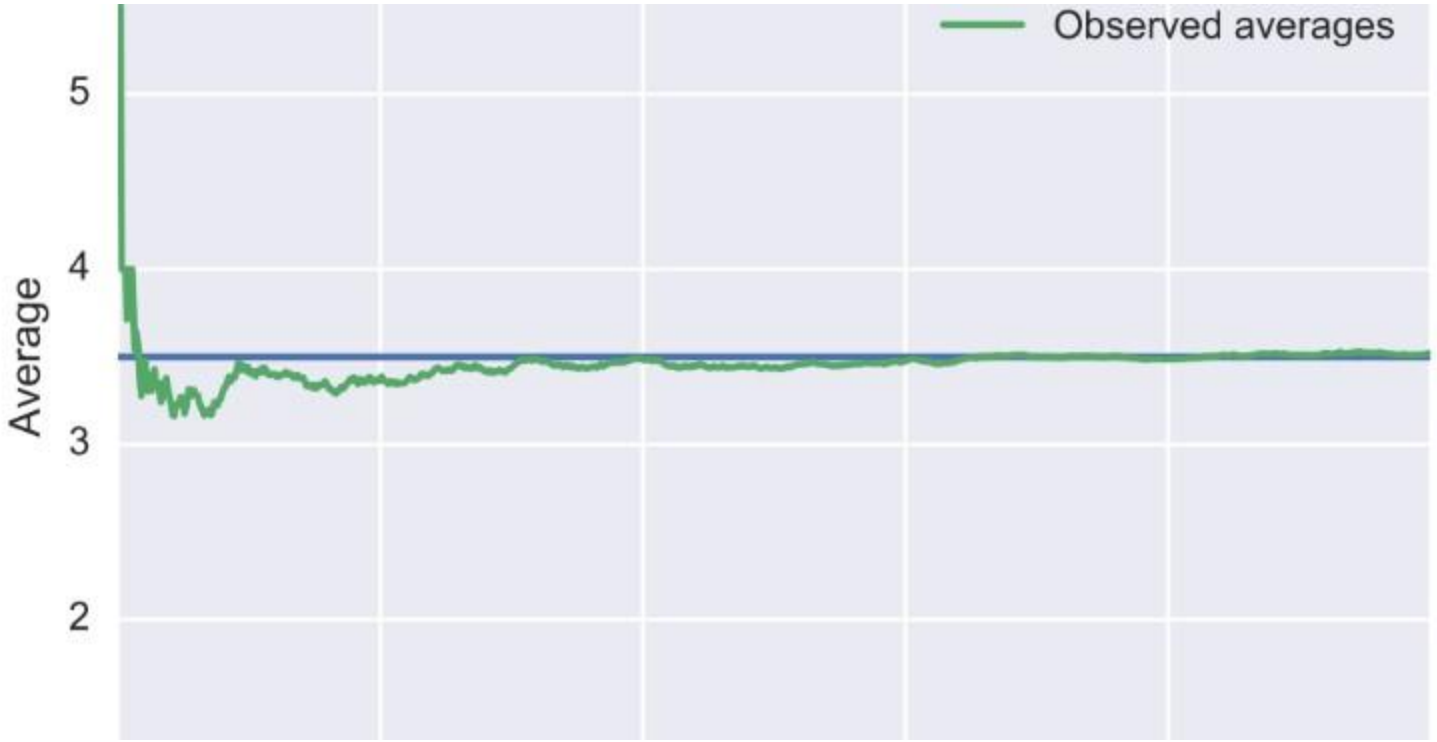


If you want to figure out the distribution of the change people carry in their pockets, using the central limit theorem and assuming your sample is large enough, you will find that the distribution is normal and bell-shaped. (credit: John Lodder)

FIGURE 7.2

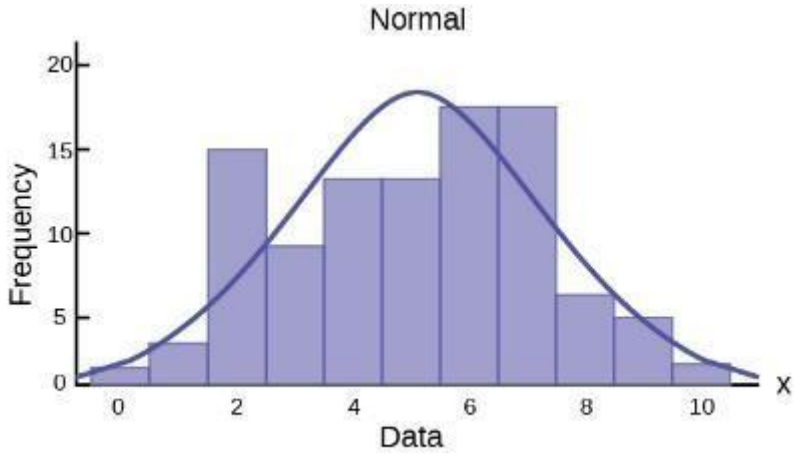


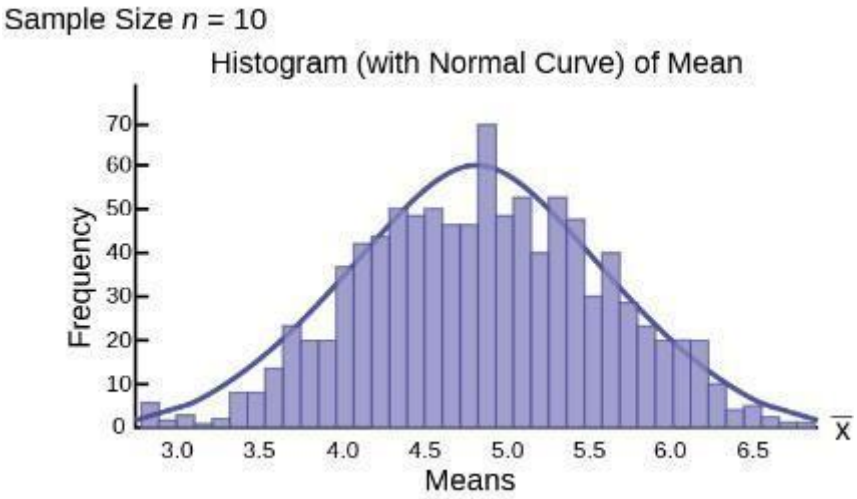
THE LAW OF LARGE NUMBERS

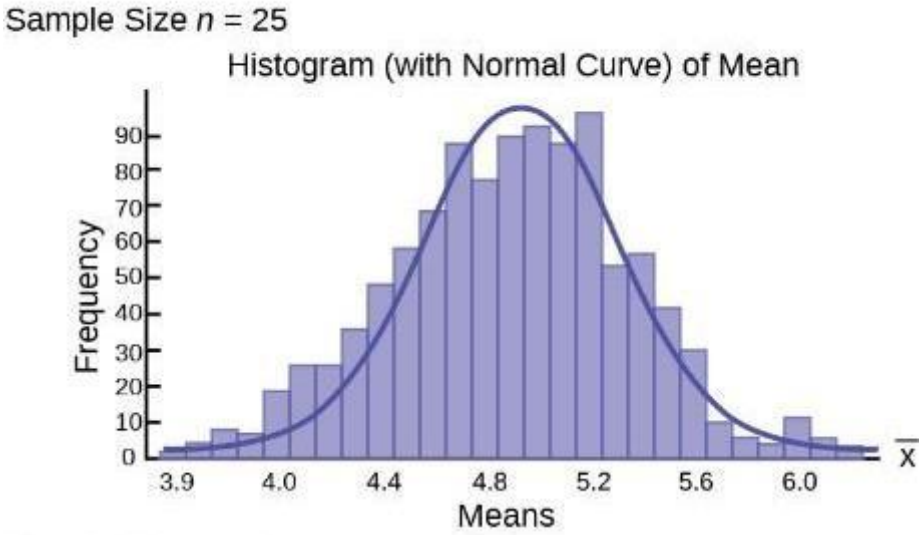


If you take samples of larger and larger size from any population, then the sampling mean of the distribution tends to get closer and closer to the true population mean.

FIGURE 7.3


Normal
Frequency
Data
X


Sample Size n = 10
Histogram (with Normal Curve) of Mean
Frequency
Means
X


Sample Size n = 25
Histogram (with Normal Curve) of Mean
Frequency
Means
X

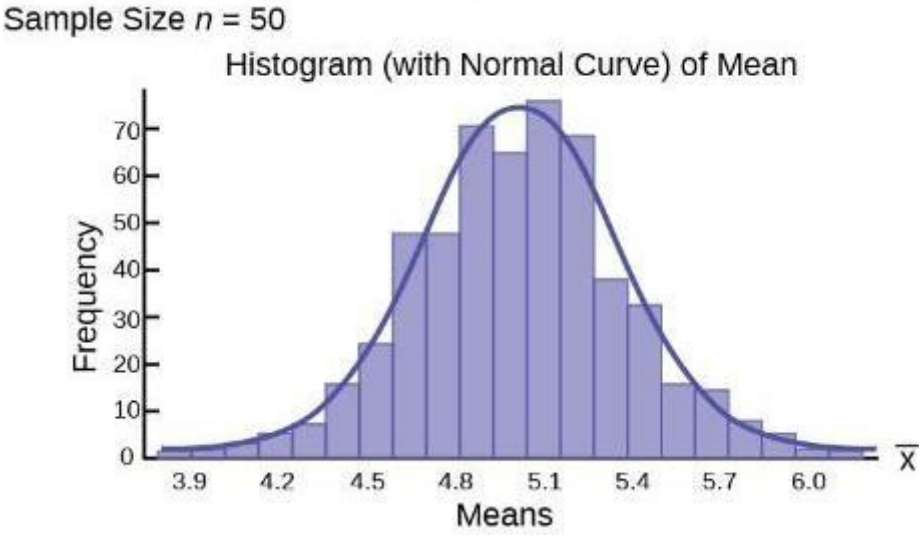

Sample Size n = 50
Histogram (with Normal Curve) of Mean
Frequency
Means
X

FIGURE 7.4

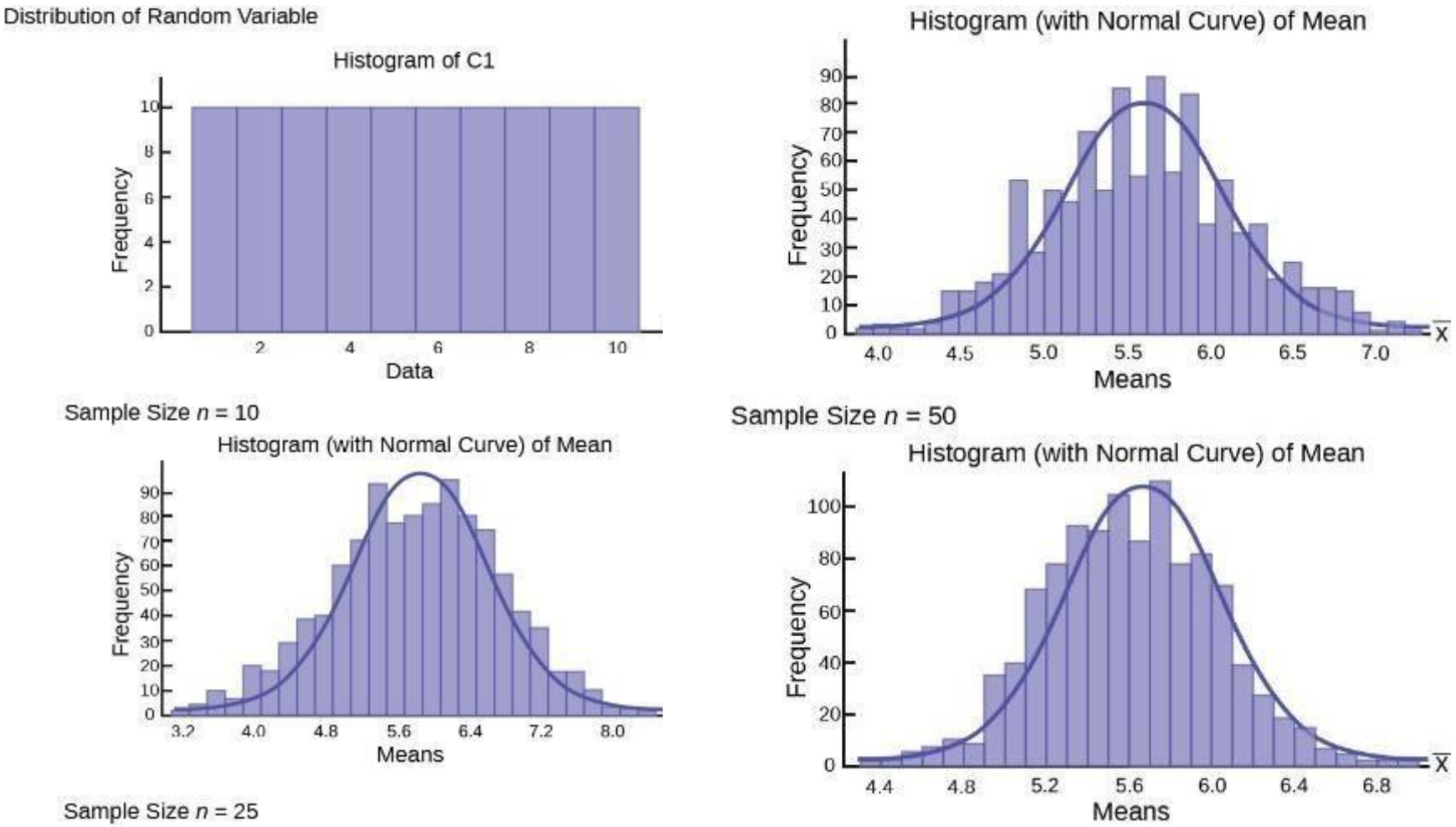
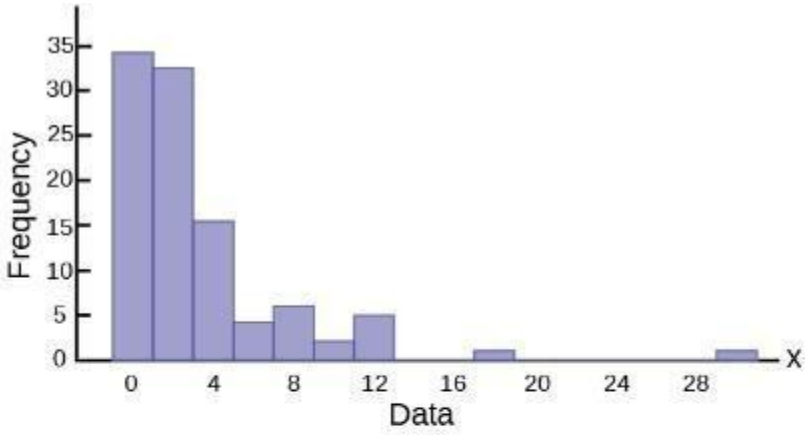
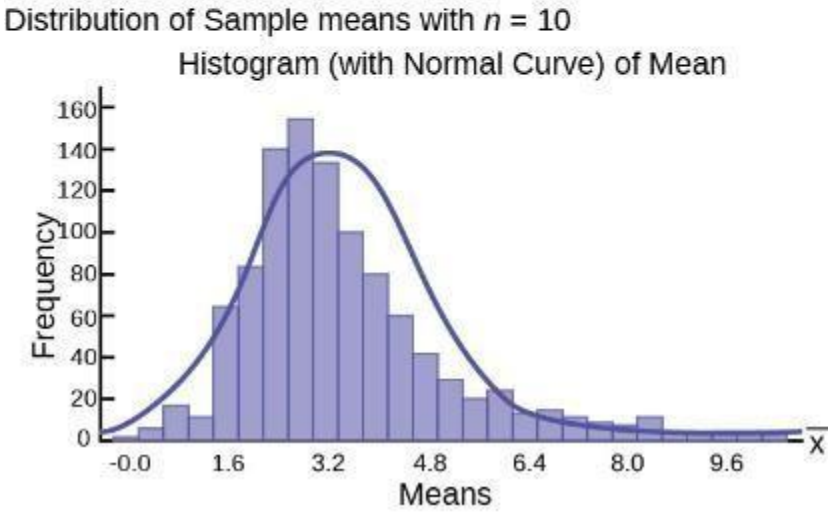
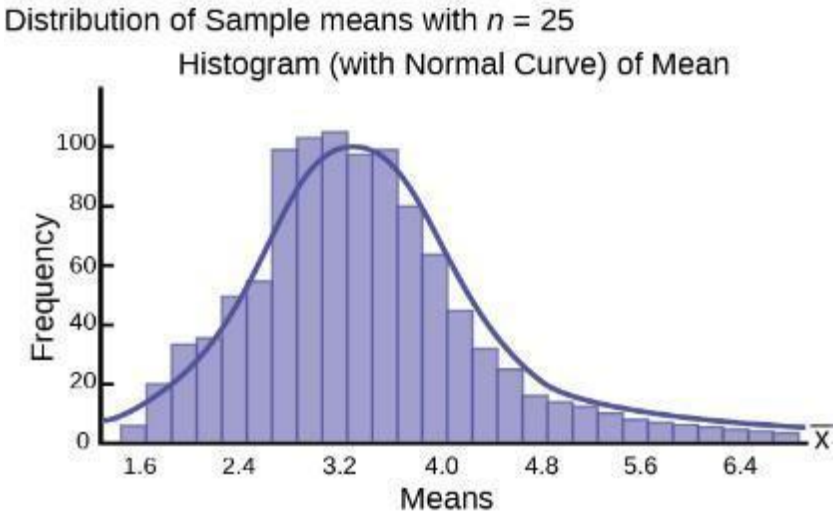
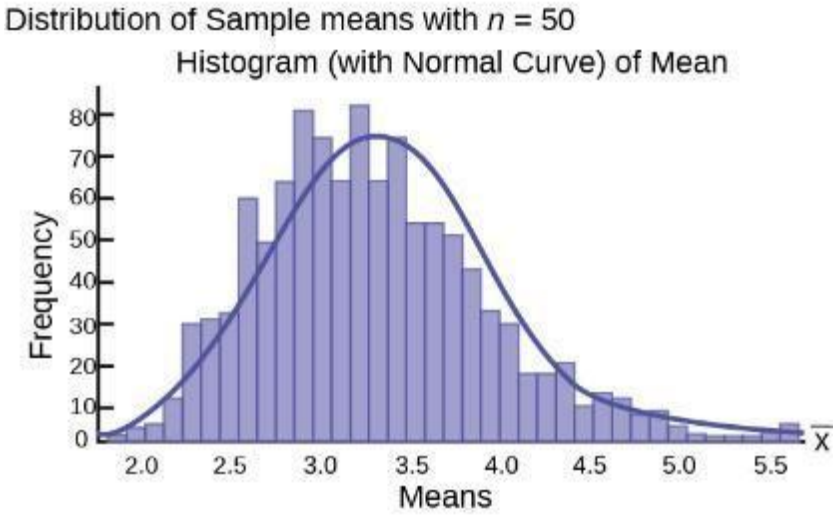

Distribution of Random Variable
Histogram of C1
Frequency
Data
Sample Size n = 10
Histogram (with Normal Curve) of Mean
Frequency
Means
Sample Size n = 25
Histogram (with Normal Curve) of Mean
Frequency
Means
Sample Size n = 50

FIGURE 7.5

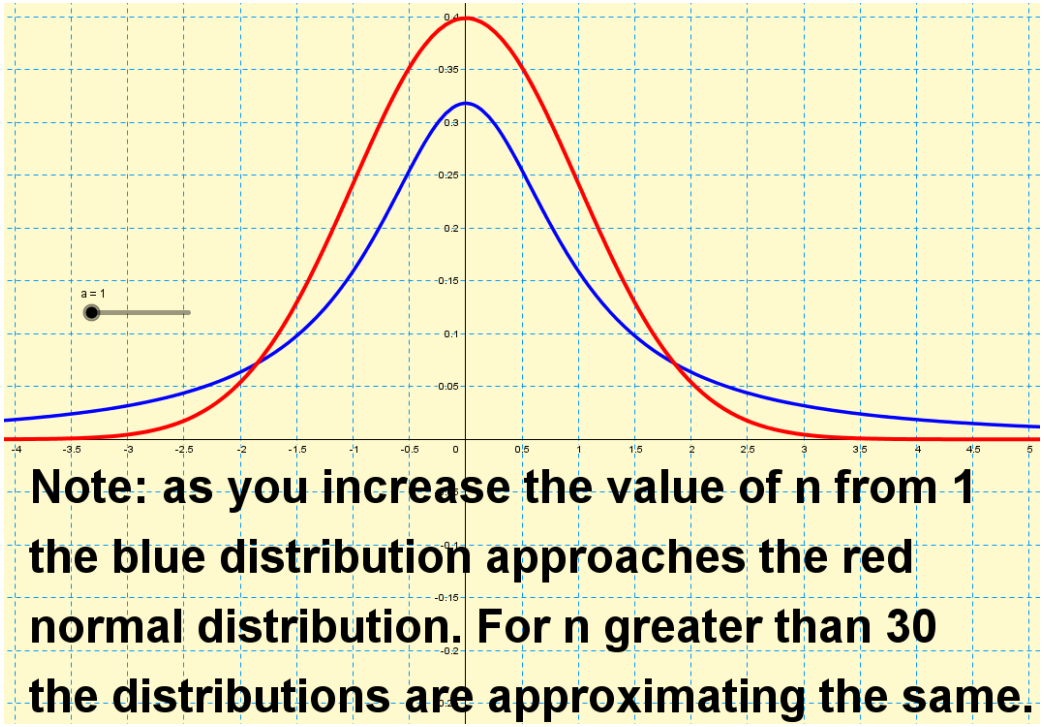

Frequency
35
30
25
20
15
10
5
0
0
4
8
12
16
20
24
28
X
Data


Distribution of Sample means with n = 10
Histogram (with Normal Curve) of Mean
Frequency
160
140
120
100
80
60
40
20
0
-0.0
1.6
3.2
4.8
6.4
8.0
9.6
X
Means


Distribution of Sample means with n = 25
Histogram (with Normal Curve) of Mean
Frequency
100
80
60
40
20
0
1.6
2.4
3.2
4.0
4.8
5.6
6.4
X
Means


Distribution of Sample means with n = 50
Histogram (with Normal Curve) of Mean
Frequency
80
70
60
50
40
30
20
10
0
2.0
2.5
3.0
3.5
4.0
4.5
5.0
5.5
X
Means

To put it more formally, if you draw random samples of size n, the distribution of the random variable $\bar{X}$, which consists of sample means, is called the sampling distribution of the mean. The sampling distribution of the mean approaches a normal distribution as n, the sample size, increases.

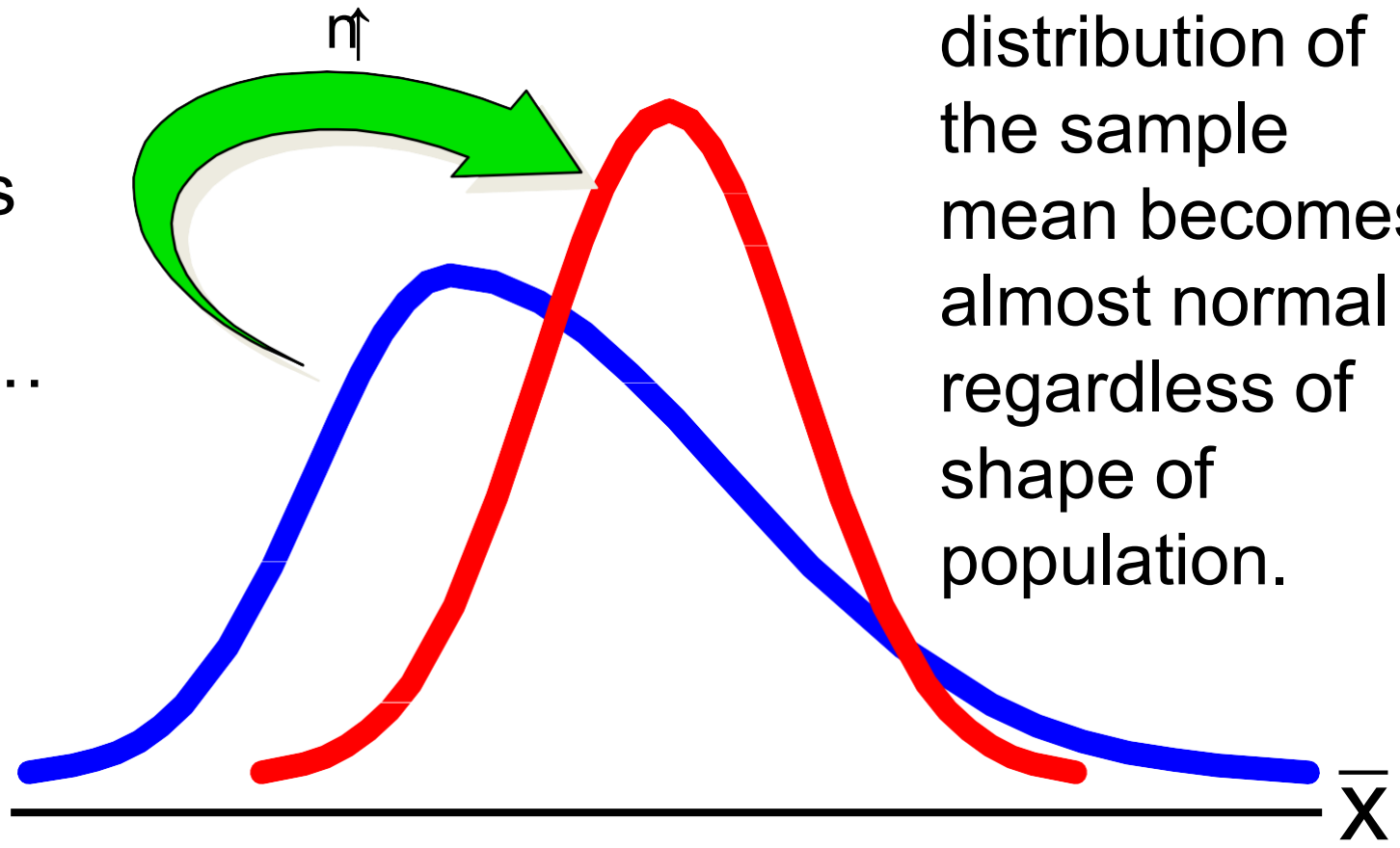


Note: as you increase the value of n from 1 the blue distribution approaches the red normal distribution. For n greater than 30 the distributions are approximating the same.

CENTRAL LIMIT THEOREM

DCOVA

As the sample size gets large enough…



the sampling distribution of the sample mean becomes almost normal regardless of shape of population.

FIGURE 7.6

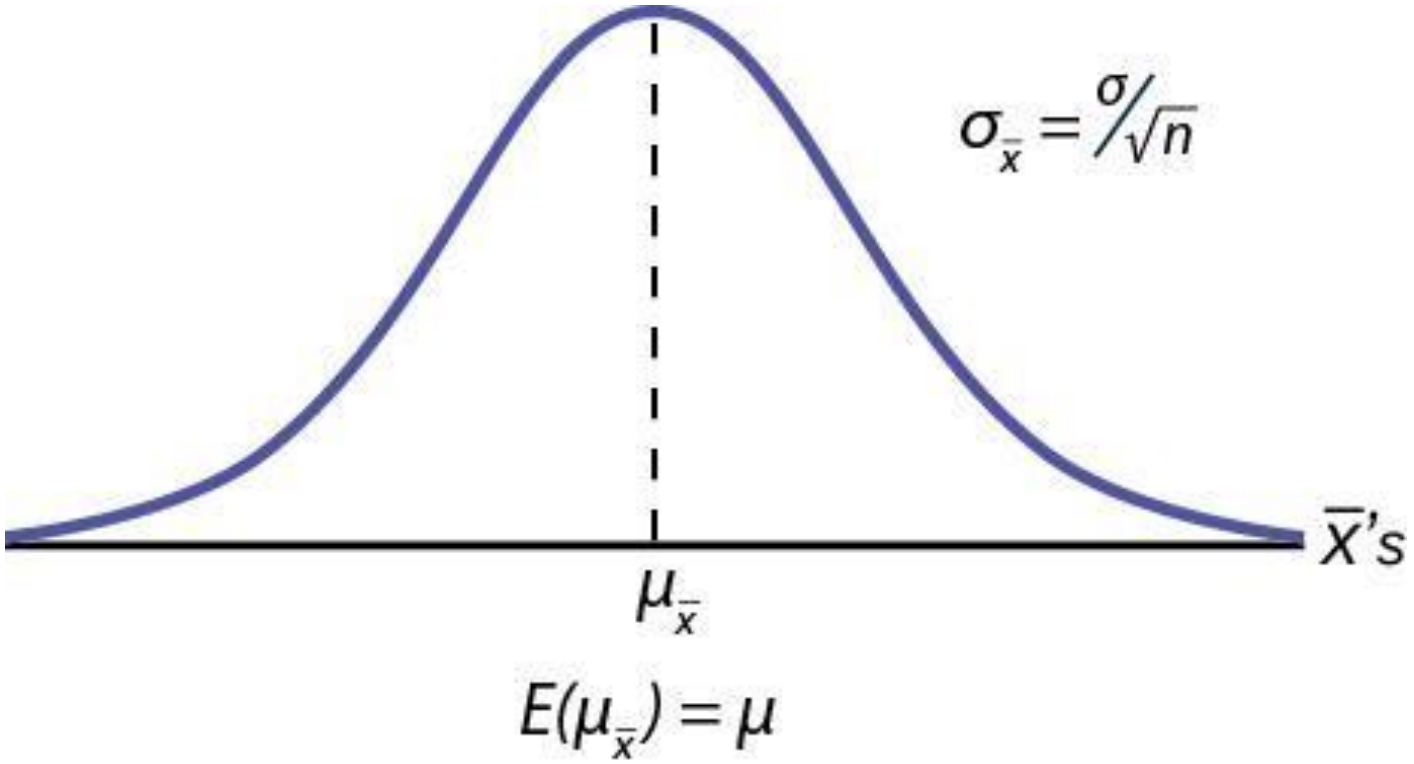


FIGURE 7.7

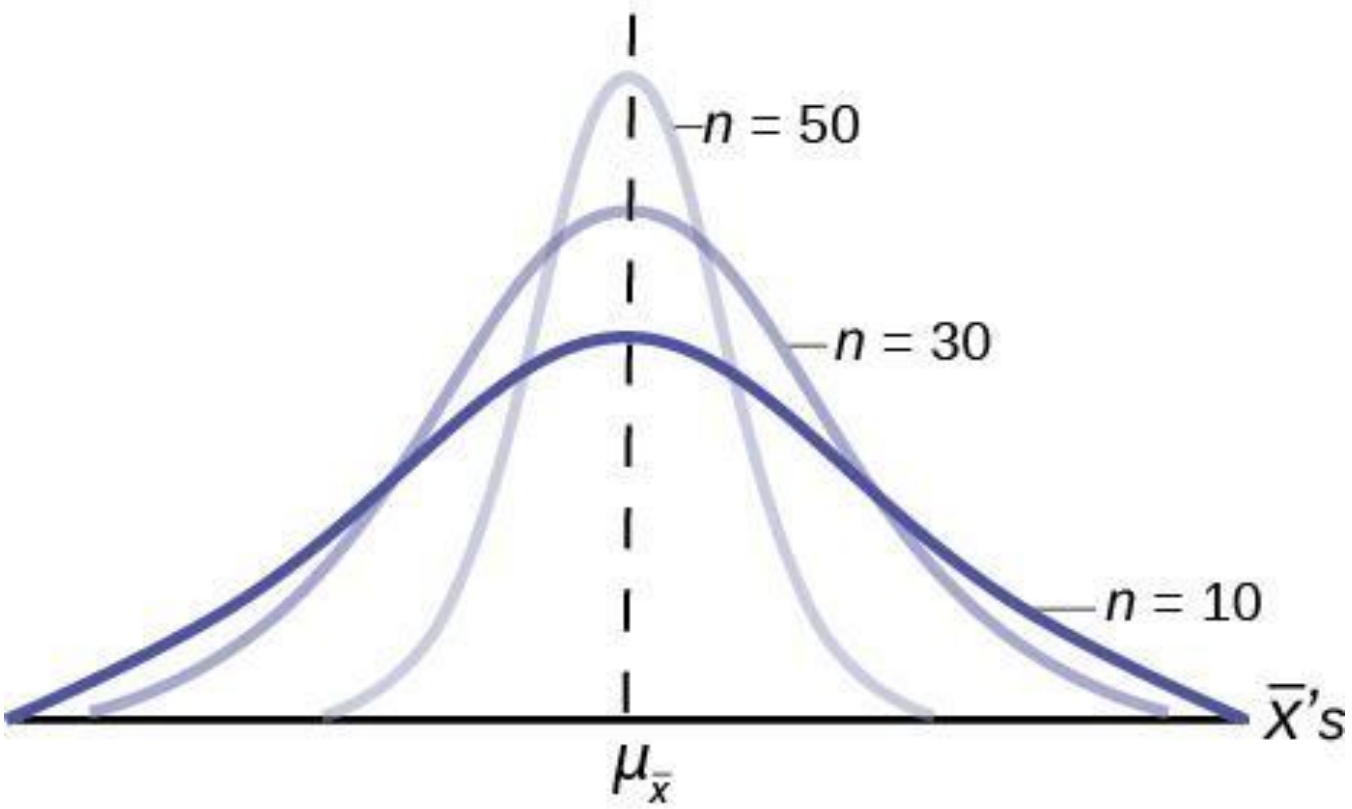
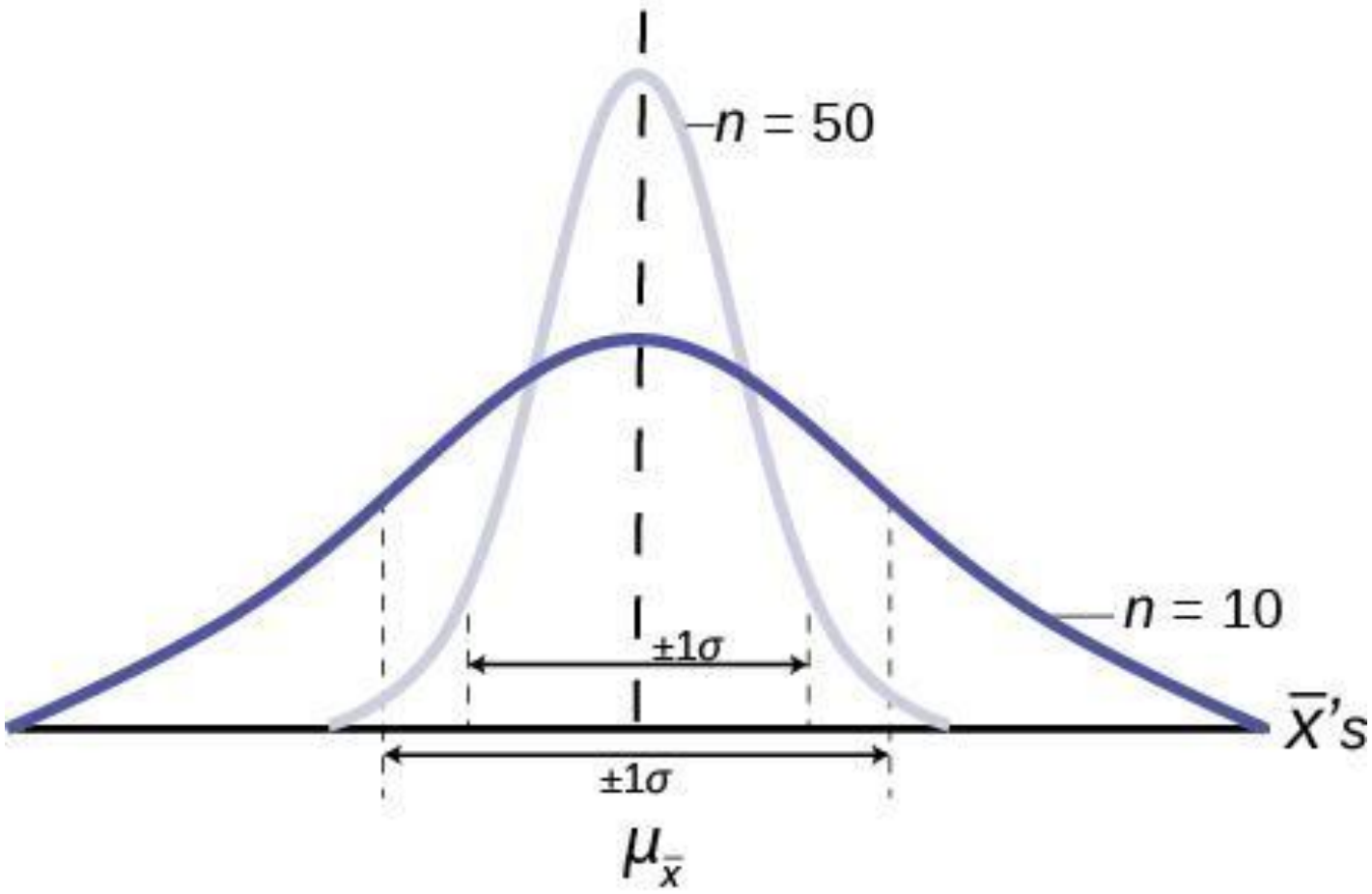


FIGURE 7.8



Example

In a recent study reported Oct. 29, 2012 on the Flurry Blog, the mean age of tablet users is 34 years. Suppose the standard deviation is 15 years. Take a sample of size n = 100.

a. What are the mean and standard deviation for the sample mean ages of tablet users?

a. Since the sample mean tends to target the population mean, we have $\mu_{\bar{X}} = \mu = 34$. The sample standard deviation is given by $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{15}{\sqrt{100}} = \frac{15}{10} = 1.5$

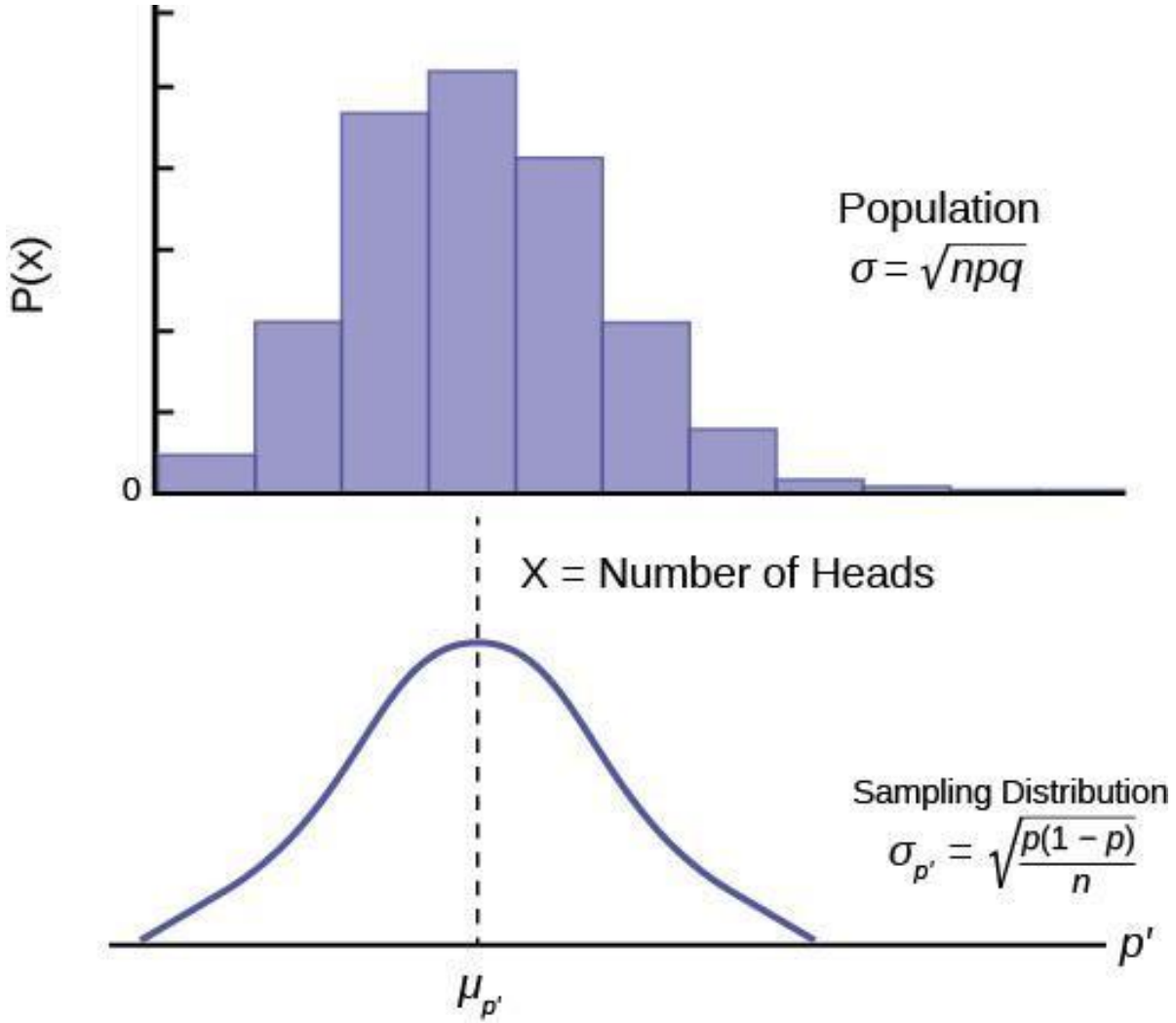
Example

In a recent study reported Oct. 29, 2012 on the Flurry Blog, the mean age of tablet users is 34 years. Suppose the standard deviation is 15 years. Take a sample of size n = 100.

b. What does the distribution look like?

b. The central limit theorem states that for large sample sizes(n), the sampling distribution will be approximately normal.

FIGURE 7.9


P(x)
0
Population
$\sigma = \sqrt{npq}$
X = Number of Heads
Sampling Distribution
$\sigma_{p'} = \sqrt{\frac{p(1-p)}{n}}$
p'
$\mu_{p'}$

This OpenStax ancillary resource is © Rice University under a CC-BY 4.0 International license; it may be reproduced or modified but must be attributed to OpenStax, Rice University and any changes must be noted.